

Transfer of Technical and Tactical Skills from Futsal to Soccer: A Systematic Review and Meta-Analysis of the ‘Donor Sport’ Concept

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Abstract

Background: The “donor sport” concept proposes that engaging in futsal can facilitate the acquisition of transferable skills for association football (soccer), consistent with ecological dynamics perspectives on constraint-led learning.

Objective: To identify, evaluate, and synthesize scientific evidence regarding the transfer of technical, tactical, and perceptual-motor skills from futsal to soccer.

Methodology: A systematic review following PRISMA 2020 guidelines (Page et al., 2021) was conducted across three electronic databases (Scopus, ScienceDirect and Web of Science Core Collection) up to November 2025. Original studies reporting quantitative data relevant to futsal-to-soccer transfer were included.

Results: Seven studies were included (N = 7). The meta-analysis revealed a very large effect size favoring futsal-based constraints for passing accuracy (SMD = 1.43; 95% CI: −0.16 to 3.03). For time-based perceptual-motor outcomes, the pooled estimate was non-significant and highly heterogeneous; field-based reactive tasks **tended to favor futsal, whereas isolated laboratory** tasks tended to favor soccer.

Conclusion: Current evidence supports the use of futsal as a potent medium to accelerate the acquisition of technical passing precision. However, the transfer of speed and agility appears to be highly dependent on context and task specificity.

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Keywords

talent development, skill acquisition, sport pedagogy, task constraints, agility

Introduction

The pursuit of elite performance in association football (soccer), a global sport characterized by continuous increases in game speed and complexity (Travassos et al., 2018; Williams et al., 2020), relies fundamentally on comprehensive talent identification and development (TID) processes (Williams et al., 2020). Elite player development increasingly advocates for early diversification of sporting experiences rather than early specialization (Fransen et al., 2012; Sieghartsleitner et al., 2018; Travassos et al., 2018). Within this context, Futsal, a 5-a-side indoor variation of soccer (Naser et al., 2017), has been postulated as an ideal “donor sport” for soccer (Oppici et al., 2019; Travassos et al., 2018; Yiannaki et al., 2018). This concept posits that Futsal’s inherent task constraints, including a smaller court size, high opponent pressure, reduced time for action, and the use of a smaller, low-bounce ball (Oppici et al., 2018b; Travassos et al., 2017) induce specific technical, tactical, and cognitive adaptations that positively transfer to the 11-a-side game (Farrow et al., 2016; Oppici et al., 2019; Travassos et al., 2018).

The viability of Futsal as a developmental tool is rooted deeply in Brazilian soccer culture (Karim et al., 2022; Marques et al., 2021), supported anecdotally by numerous world-class Brazilian soccer players, such as Pelé, Ronaldinho, and Neymar, who attribute their superior technical and tactical proficiency to early experience in the indoor format (Ferreira et al., 2025; Sá et al., 2010; Travassos et al., 2018). This practice environment compels players to develop superior technical control and precision, particularly in passing and dribbling within confined spaces (Benvenuti et al., 2010; Oppici et al., 2018a; Silva et al., 2010). Futsal demands quick and effective responses in unpredictable situations (Corrêa et al., 2012; Garganta, 2002; Hebert & Trichês dos Santos, 2009), thereby enhancing faster decision-making and reactive agility (Benvenuti et al., 2010; Naser et al., 2017). Comparative studies show that Futsal players perform significantly better in reactive agility tests (RVS-T) and demonstrate lower Decision-Making Time (DMT) compared to soccer players, suggesting a sport-specific effect of this high-intensity, space-constrained practice.

Optimal talent acquisition is often described through the framework of the Developmental Model of Sport Participation (DMSP) (Côté, 1999) or the concept of Specialized Diversification (Sieghartsleitner et al., 2018). Developmental activities are commonly differentiated into deliberate practice (structured training) and deliberate play (unstructured, high-satisfaction play) (Côté et al., 2007; Ericsson et al., 1993). Research on Brazilian professional female soccer players found that a pathway combining structured activities in soccer and Futsal (deliberate practice) with unstructured play (deliberate play) during childhood (6–12 years) and early adolescence (13–15 years) is positively associated with both the quality and speed of decision-making skills in adulthood (Machado et al., 2023,

2024). Specifically, players with higher cognitive performance accumulated more hours of deliberate practice in futsal and a higher total practice time during early adolescence (13–15 years) compared to low-skill players (Machado et al., 2024).

Despite the strong theoretical and anecdotal rationale, the quantitative empirical evidence supporting a direct and reliable skill transfer between Futsal and soccer remains limited and characterized by substantial methodological heterogeneity. Previous studies have shown inconsistent results across different skill modalities. While the concept of Futsal as a donor sport is conceptually strong due to the inherent similarities between the sports (Travassos et al., 2018; Yiannaki et al., 2018), certain findings point toward task specificity:

- (1) **Technical Skills:** Multiple intervention studies indicate that practicing passing with Futsal constraints (e.g., the low-bounce ball) results in a very large positive effect size ($SMD = 1.43$) for passing accuracy in soccer environments, accelerating skill acquisition even in novices (Oppici et al., 2018b).
- (2) **Perceptual-Motor Skills:** The transfer of speed-related skills appears dependent on the task environment. While Futsal players show superior performance in reactive tasks embedded in ecological contexts (Benvenuti et al., 2010), their advantage often disappears or reverses in isolated laboratory assessments of reaction time (Andrew & Causer, 2022; Silva et al., 2022).
- (3) **Dribbling/Kinematics:** Studies analyzing dribbling reveal that successful performance in Futsal is strongly influenced by coordination aspects like smaller interpersonal distances and higher dribbling velocities, a finding that does not translate to soccer (Silva et al., 2025). Conversely, success in soccer dribbling, as opposed to futsal, appears to be related to the defender's Reaction Time (RT). This suggests that distinct constraints in each domain cultivate specialized solutions for offensive play.

This existing body of fragmented evidence, coupled with variability in design (ranging from correlational/retrospective to quasi-experimental and isolated RCTs), necessitates a critical and quantitative synthesis. A meta-analysis is essential to rigorously evaluate the magnitude of the transfer effect across key dimensions of performance and to provide robust conclusions for high-level athletic development programs.

Theoretical framework and operational definition of transfer. The 'donor sport' concept originates within the Athletic Skills Model (ASM) which proposes that exposure to related sports can 'donate' functional movement and perceptual-cognitive tendencies to a target sport by shaping athletes' adaptive capacities through constraints-led experiences and ecological dynamics. In the ASM, donor sports are expected to enrich athletes' perceptual-motor landscapes by (i) developing broad coordinative foundations via a concentric approach, (ii) exploiting transferability through multisport/donor-sport engagement, and (iii) creating challenging environments that promote adaptability and perception-action coupling (Savelsbergh & Wormhoudt, 2021). Accordingly, in the present review we operationalized transfer as evidence that futsal exposure (training interventions or developmental history) is associated with

measurable changes in soccer-relevant performance outcomes; comparative futsal-versus-soccer studies were treated as plausibility evidence of constraint-driven adaptations rather than direct causal transfer.

The objective of this systematic review and meta-analysis is to identify, evaluate, and quantitatively synthesize the extant scientific evidence regarding the transfer of technical, tactical, and perceptual-motor skills from Futsal to association soccer, thereby empirically testing the validity of the “donor sport” concept. This effort seeks to provide definitive, quantifiable evidence to guide coaches, clubs, and sports scientists in constructing evidence-based youth talent development pathways.

Methodology

This systematic review and meta-analysis were conducted and reported following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al., 2021). The review protocol was designed and registered previously to ensure transparency and reproducibility.

Based on the PICO(S) format (Population, Intervention/Exposure, Comparison, Outcome, and Study Design), to ensure a rigorous study selection process, the following inclusion and exclusion criteria were established: (1) Soccer (football) and/or futsal players, with no restrictions on age, gender, or competitive level (youth, elite, professional), provided they had no active pathologies affecting performance; (2) Studies that analyzed the implementation of futsal training programs in soccer players or the effect of a historical background in futsal practice; (3) The comparison was established between soccer players with futsal experience versus those without (soccer-only), or between an intervention group (futsal) and a control group (traditional soccer training); (4) Studies were required to report quantitative data on at least one primary outcome related to: (a) Technical skills (e.g., passing accuracy/speed, ball control, dribbling), (b) Tactical/Cognitive behavior (e.g., decision-making, visual search behavior/scanning, anticipation), or (c) Physical-motor skills (e.g., reactive agility, change of direction speed) transferable between sports; and (5) Study Design: Original research articles of randomized controlled trials (RCTs), quasi-experimental, and comparative cross-sectional designs were included. In turn, the exclusion criteria were as follows: (1) Studies focused exclusively on sport management, economy, sociology, or clinical psychology unrelated to performance; (2) Purely epidemiological or medical studies (e.g., injury rates, hormonal profiles) lacking performance or skill testing; (3) Studies on non-specific “Small-Sided Games” (SSGs) that did not utilize specific futsal rules or characteristics (e.g., low-bounce ball, hard surface); (4) Document types such as conference abstracts, letters to the editor, books, unpublished theses, or opinion pieces, systematic reviews, scoping reviews, and meta-analyses were excluded as eligible study designs; however, their reference lists were manually inspected to identify potentially relevant primary studies; and (5) Studies involving unrelated sports (e.g., American Football, Gaelic Football, Rugby).

Information Sources and Search Strategy

A systematic literature search was conducted in three electronic databases: Scopus, ScienceDirect and Web of Science (Core Collection). The search was completed in November 2025 and included all articles published up to that date. We included studies published in English, Spanish, and Portuguese because these languages cover a substantial proportion of futsal and soccer development research, and they were the languages available for full-text appraisal within the review team. We acknowledge that language restrictions may introduce language bias; therefore, language was applied as an eligibility criterion (not as a search filter when possible) and is discussed as a limitation.

The search strategy was designed to maximize the identification of relevant literature by combining keywords using Boolean operators (AND, OR) and truncations: (a) Block 1 (Futsal): (“futsal” OR “indoor soccer” OR “five-a-side”); (b) Block 2 (Target Sport): (“football” OR “soccer” OR “association football”); (c) Block 3 (Topic/Outcome): (“transfer” OR “transition” OR “skill*” OR “talent” OR “acquisition” OR “decision making” OR “perceptual” OR “motor” OR “development”).

Study Selection Process

Search results were exported to the Mendeley reference manager for organization and the removal of duplicate records. Title/abstract and full-text screening were independently performed by two reviewers. Disagreements were resolved by consensus; when needed, a third reviewer arbitrated. The process consisted of the following phases: (1) Identification and De-duplication: Initially, records identified from the electronic databases were collated, and duplicate entries were removed to ensure a unique dataset for screening; (2) Screening by Title: The first screening phase involved assessing the titles of the non-duplicate records. At this stage, documents were excluded if they were clearly non-original research, or if they addressed irrelevant contexts or topics outside the scope of sports performance; (3) Screening by Abstract: The remaining records underwent a secondary screening based on their abstracts. Studies were excluded in this phase if they failed to establish a direct relationship, comparison, or transfer link between futsal and soccer, or if they focused exclusively on variables not pertinent to the study’s objective; (4) Eligibility (Full-Text Assessment): The full texts of all reports deemed potentially relevant were retrieved and assessed in detail against the eligibility criteria. During this final phase, studies were excluded if they presented incorrect or deficient study designs, lacked an explicit analysis of transfer or comparison between the two sports, maintained an exclusive medical focus, or utilized instruments or samples that were invalid for data extraction and meta-analysis. The author critically evaluated the full text of each candidate study to reach a definitive inclusion decision, explicitly documenting the specific grounds for exclusion for transparency. Given that the screening process was conducted by two reviewers, measures of inter-rater reliability could not be calculated; this methodological constraint and its potential implications are further addressed in the Limitations section.

Data Extraction Process

A standardized extraction form was used to retrieve data from the included studies. The extracted data included: (a) Bibliographic information (Author, year); (b) Population characteristics (Sample size, age, gender, level); (c) Intervention details (Duration, specific futsal constraints used); (d) Measurement methodology (Tests and instruments used); and (e) Main results (Means and Standard Deviations for the outcomes of interest).

Methodological Quality Assessment

The methodological quality of the included studies was assessed using the Downs and Black (D&B) checklist. This tool was selected for its suitability in evaluating both randomized and non-randomized studies, which aligns with the heterogeneous nature of the included literature (cross-sectional, quasi-experimental, and RCTs). The scale evaluates 27 items across four domains: Reporting, External Validity, Internal Validity (Bias and Confounding), and Statistical Power. A numeric score was assigned to each study to provide an objective assessment of its quality and presented in the visual summary is provided in the Results ([Table 2](#)).

Statistical Analysis and Meta-Analysis

For the quantitative synthesis, statistical analyses were performed using RStudio software (v. 2023.12) with the meta package. The Standardized Mean Difference (SMD), adjusted as Hedges' g to correct for small sample size bias, was calculated for continuous variables. Model: A Random Effects Model was applied using the Restricted Maximum Likelihood (REML) estimator, assuming that the true effect size varies between studies due to the heterogeneity of the experimental designs. Heterogeneity: Heterogeneity was quantified using the I^2 statistic (where values $>50\%$ indicate substantial heterogeneity) and Cochran's χ^2 test. Interpretation: The magnitude of the effect size was classified according to Cohen's guidelines: trivial (<0.2), small ($0.2-0.5$), moderate ($0.5-0.8$), and large (>0.8). Given the small number of studies contributing to some syntheses, pooled estimates were interpreted cautiously and, where applicable, more conservative random-effects inference approaches suitable for few studies (e.g., Hartung–Knapp-type adjustments) were considered ([Röver, C., et al., 2015](#)).

Results

Selection and Characteristics of Studies

The study selection process is summarized in the PRISMA flow diagram ([Figure 1](#)). The initial systematic search across the electronic databases (Scopus, ScienceDirect, and Web of Science) yielded a total of 780 records. After exporting these results to the reference manager and removing 97 duplicates, a total of 683 unique records proceeded to the title screening phase. During the initial screening by title, 468 articles were

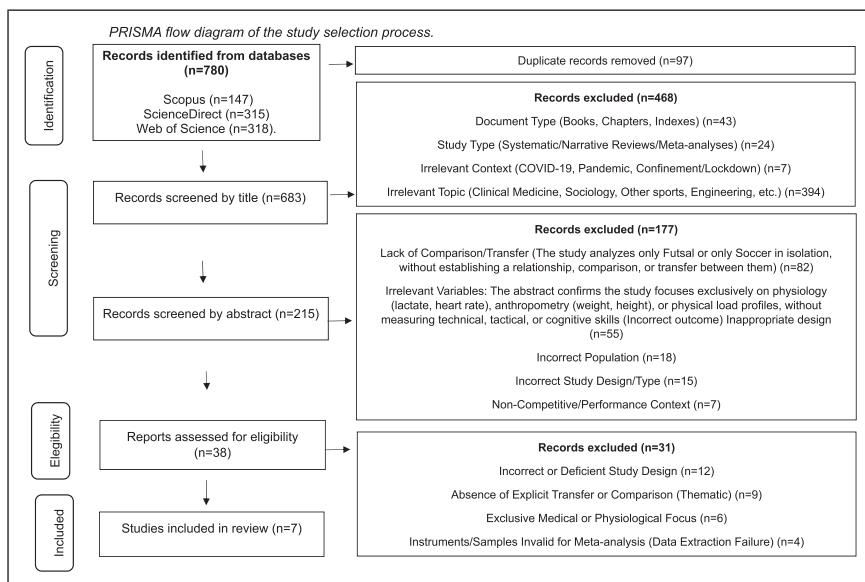


Figure 1. PRISMA flow diagram of the study selection process

discarded. The primary reasons for exclusion at this stage were: irrelevant topics unrelated to sports performance ($n = 394$; e.g., clinical medicine, sociology, or engineering), specific document types such as books or book chapters ($n = 43$), study types including reviews or meta-analyses ($n = 24$), or irrelevant contexts such as the COVID-19 pandemic ($n = 7$).

Subsequently, the abstracts of the remaining 215 records were assessed. A total of 177 records were excluded, mainly due to a lack of explicit comparison or transfer analysis between futsal and soccer ($n = 82$), or the focus on irrelevant variables ($n = 55$; e.g., studies focused exclusively on physiology or anthropometry without technical-tactical measurement). Other reasons included incorrect study populations ($n = 18$), inappropriate study design ($n = 15$), or non-competitive contexts ($n = 7$). Consequently, 38 reports were retrieved for full-text eligibility assessment. Upon detailed examination, 31 studies were excluded. The specific grounds for exclusion at this final stage were: incorrect or deficient study design ($n = 12$), absence of explicit thematic transfer or comparison ($n = 9$), an exclusive medical or physiological focus ($n = 6$), or invalid instruments/samples that prevented successful data extraction for meta-analysis ($n = 4$). Finally, 7 studies satisfied all eligibility criteria and were included in the quantitative and qualitative synthesis.

Consequently, a total of seven original research articles were deemed eligible for inclusion in the systematic review and meta-analysis. These studies satisfied the strict thematic criteria by explicitly investigating the transfer or comparison of technical, tactical, and perceptual-cognitive skills between futsal and association soccer. The

Table 1. Characteristics of the Included Studies

Author(s) (Year)	Population and sample (N)	Study design	Intervention/Exposure details	Outcome variables	Relevant main findings
Benvenuti et al. (2010)	36 female players (18 elite futsal, 18 elite soccer).	Comparative cross-sectional	Comparison of perceptual- motor performance between elite futsal and soccer players in specific field tests.	Reactive agility (RVST), planned agility (PVS-T), decision making time (DMT).	Futsal players were significantly faster in reactive agility and had lower decision making time compared to soccer players. No differences found in planned agility.
Caglayan et al. (2018)	48 youth soccer players (U13- U14).	Quasi- experimental (Pre-Post)	8-week training program. Exp. Group used a futsal ball; control group used a standard soccer ball.	Passing accuracy (Mor- Christian test), dribbling speed.	The group training with the futsal ball showed significantly greater improvement in passing accuracy compared to the control group. No significant difference in dribbling speed improvement.
Oppici et al. (2018a)	24 youth players (12 futsal background, 12 soccer only).	Quasi- experimental (transfer task)	Performance assessment under futsal task constraints vs. Soccer task constraints.	Passing accuracy (error magnitude), orientation of attention (head turn).	Futsal players showed positive transfer of passing skills to soccer constraints. Futsal constraints facilitated better attention orientation in all players.
Oppici et al. (2018b)	40 novice players (adults).	Randomized controlled trial (RCT)	Training intervention using futsal ball (low restitution) vs. Soccer ball (high restitution) to learn passing skills.	Passing accuracy, passing consistency.	Training with a futsal ball accelerated the acquisition of passing skills, which successfully transferred to performance with a standard soccer ball.

(continued)

Table 1. (continued)

Author(s) (Year)	Population and sample (N)	Study design	Intervention/Exposure details	Outcome variables	Relevant main findings
Silva et al. (2022)	92 total actions analyzed (futsal vs. Soccer players).	Cross-sectional	Comparison of dribbling success based on defender's reaction time (RT) and anticipation in 1v1 duels.	Dribbling success rate, reaction time (RT), anticipation frequency.	In soccer, dribbling success was dependent on the defender's RT. In futsal, neither RT nor anticipation predicted success, suggesting higher complexity/ other factors involved.
Machado et al. (2024)	50 professional female soccer players.	Observational (retrospective)	Retrospective analysis of deliberate practice hours in futsal and soccer during developmental stages.	Decision-making (DM) quality and response time (TacticUP video test).	Accumulated hours of futsal practice in childhood (7-12 yrs) correlated significantly with better tactical decision-making quality in adulthood.
Silva et al. (2025)	NA (sample of 1v1 duels in futsal and soccer).	Cross-sectional	Analysis of attacker-defender interaction dynamics in 1v1 situations.	Interpersonal distance, relative velocity, dribbling outcome.	Spatiotemporal coordination (distance/velocity) significantly affected dribbling outcomes in futsal but not in soccer, indicating higher constraint specificity in futsal.

selected body of evidence comprises a heterogeneous mix of study designs, including cross-sectional analyses, quasi-experimental studies, and randomized controlled trials, offering a broad perspective on the ‘donor sport’ concept. A comprehensive overview of the methodological characteristics, sample details, intervention protocols, and key findings of each included study is provided in [Table 1](#).

Characteristics of the Included Studies

The seven included studies were published between 2010 and 2025, reflecting a persistent research interest in the transfer of skills between small-sided sports and association soccer over the last 15 years. Geographically, the research encompasses a diverse international context, with contributions from Europe (Italy, Turkey), South America (Brazil), and Oceania (Australia), highlighting the global relevance of the ‘donor sport’ concept.

The study populations exhibited significant heterogeneity regarding gender, developmental stage, and competitive level. Participants ranged from youth categories (U13–U14) ([Caglayan et al., 2018](#); [Oppici et al., 2018a](#)) and adult novices ([Oppici et al., 2018b](#)) to high-performance cohorts, including elite female players ([Benvenuti et al., 2010](#)) and professional female athletes ([Machado et al., 2024](#)). Sample sizes varied considerably depending on the study design, ranging from smaller experimental cohorts of $N = 24$ ([Oppici et al., 2018a](#)) to larger retrospective samples of $N = 50$ ([Machado et al., 2024](#)) and specific action-based analyses involving $N = 92$ duel situations ([Silva et al., 2022](#)).

Methodologically, the review comprises a robust mix of study designs, including comparative cross-sectional studies, quasi-experimental interventions, and a randomized controlled trial (RCT). The primary outcomes focused on the quantitative analysis of technical proficiency (specifically passing accuracy and dribbling), perceptual-cognitive skills (decision-making time, anticipation, and attention orientation), and motor capabilities (reactive agility). These variables were assessed using a combination of validated field tests (e.g., Mor-Christian Football Aptitude Test, Reactive Agility Test), video-based tactical assessments (e.g., TacticUP), and kinematic analysis of spatiotemporal constraints.

Methodological Quality Assessment

The methodological quality of the included studies, assessed via the Downs and Black checklist, ranged from 16 to 23 points out of a maximum of 27, indicating a heterogeneity in quality spanning from ‘Moderate’ to ‘High’. The randomized controlled trial conducted by [Oppici et al. \(2018b\)](#) achieved the highest rating (23/27), distinguishing itself through robust internal validity measures, including partial blinding and a priori power analysis.

The remaining six studies were classified within the ‘Moderate’ to ‘Moderate-High’ quality range (scores 16–19). The primary methodological constraints identified in these cross-sectional and quasi-experimental designs were the absence of randomization, which introduces potential selection bias, and the inherent difficulty of controlling for confounding variables in ecological sports settings. Despite these limitations, the reporting quality across

Table 2. Methodological Quality Assessment of the Included Studies Using the Downs and Black Checklist

N.º	Study Title (Short Reference)	Main Study Design	Estimated Score (Max 27 pts)	Final Rating
1	Agility assessment in female futsal and soccer players (Benvenuti et al., 2010)	Comparative cross-sectional (observational)	17/27	Moderate quality
2	Futsal task constraints promote transfer of passing skill to soccer task constraints (Oppici et al., 2018a)	Quasi-experimental (Pre-existing groups)	18/27	Moderate-high quality
3	Contribution of deliberate practice, play, and futsal... (Machado et al., 2024)	Observational (retrospective, correlational)	19/27	Moderate-high quality
4	Success and failure of dribbling in futsal and football based on reaction time and anticipation (Silva et al., 2022)	Cross-sectional (observational)	17/27	Moderate quality
5	The attacker-defender's spatiotemporal relationship affects the dribbling outcome in futsal but not in soccer (Silva et al., 2025)	Cross-sectional (observational)	16/27	Moderate quality
6	The effects of trainings with futsal ball on dribbling and passing skills on youth soccer players (Caglayan et al., 2018)	Quasi-experimental (intervention)	17/27	Moderate quality
7	The influence of a modified ball on transfer of passing skill in soccer (Oppici et al., 2018b)	Randomized controlled trial (RCT)	23/27	High quality

studies was generally sufficient to allow for data extraction. A visual summary of the risk of bias assessment and individual ratings is presented in Table 2.

Meta-Analysis Results

Transfer to Passing Accuracy (Technical Skill). A quantitative synthesis was performed on three studies that evaluated the effects of futsal-specific constraints (e.g., low-bounce ball, reduced space) on passing accuracy in soccer players. The meta-analysis revealed a very large overall effect size (SMD = 1.43; 95% CI: -0.16 to 3.03) favoring futsal-based interventions. Although substantial statistical heterogeneity was observed ($I^2 = 86.9\%$, $p < .001$), the direction of the effect was consistently positive across all individual studies. Notably, the investigations with the highest level of experimental control (Oppici et al.) reported effect sizes exceeding 2.0, suggesting an acute and

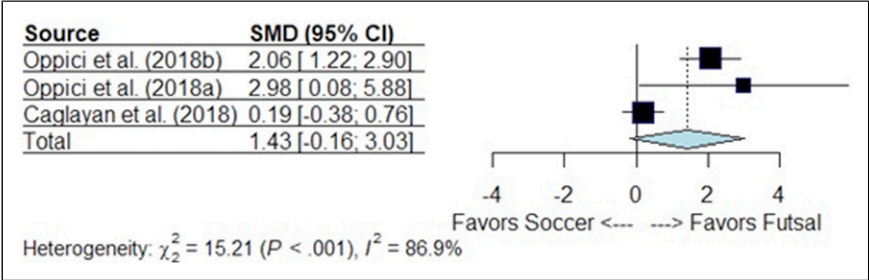


Figure 2. Forest plot of the standardized mean differences (SMD) for time-based perceptual-motor tasks

Note: Positive values (right side) indicate faster performance favoring futsal players, while negative values (left side) indicate faster performance favoring soccer players. The squares represent individual study effects, and the horizontal lines indicate the 95% confidence intervals (CI)

significant transfer of technical proficiency. The forest plot summarizing these findings is presented in [Figure 2](#).

Perceptual-Motor Skills and Speed (Time-Based Tasks). The meta-analysis of time-dependent variables—encompassing agility, dribbling speed, and reaction time—incorporated four distinct comparisons. Due to conceptual and methodological heterogeneity across time-based outcomes (e.g., reactive agility in representative field tests vs isolated laboratory reaction-time tasks), we did not interpret a single pooled estimate as evidence of a unified ‘transfer effect’. Instead, we structured interpretation by task representativeness (field-based vs laboratory-based assessments).

However, a distinct dichotomy emerged when examining the ecological validity of the assessment protocols:

- Field-based assessments: Studies utilizing simulated game scenarios ([Benvenuti et al., 2010](#)) demonstrated large effect sizes favoring futsal players in reactive agility and decision-making speed ($SMD > 1.50$), suggesting a transfer of cognitive speed in representative environments.
- Laboratory or isolated tasks: Conversely, investigations employing video-based reaction tests or linear dribbling drills without opposition ([Caglayan et al., 2018](#); [Silva et al., 2022](#)) indicated superior or comparable performance by soccer players, shifting the effect direction toward the control group.

The forest plot summarizing these divergent findings is presented in [Figure 3](#).

Synthesis of Findings

Seven studies ($N = 7$) met the eligibility criteria for the systematic review and were included in the quantitative and qualitative synthesis. These studies adopted varied

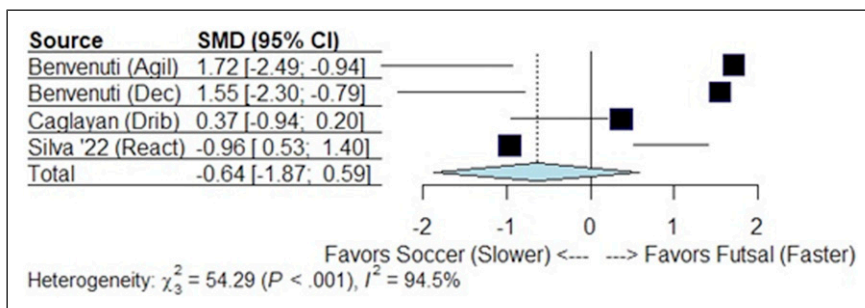


Figure 3. Forest plot of the standardized mean differences (SMD) for passing accuracy between futsal and soccer training. The squares represent individual study effects, and the horizontal lines indicate the 95% confidence intervals (CI). The diamond represents the overall pooled effect size

designs, including Randomized Controlled Trials (RCTs), quasi-experimental interventions, comparative cross-sectional analyses, and retrospective correlational designs, ranging in methodological quality from moderate (Benvenuti et al., 2010; Caglayan et al., 2018; Silva et al., 2022, 2025) to high quality (Oppici et al., 2018b).

The synthesis of findings is organized by the domain of performance evaluated: Technical Skills, Perceptual-Motor/Cognitive Skills, and Contextual/Biomechanical Specificity.

Technical Proficiency

The quantitative synthesis consistently supports the concept that practicing under Futsal-specific constraints enhances technical skills, particularly passing accuracy, with a pronounced effect size.

- **Passing Accuracy (Transferable Skill):** Intervention studies demonstrated that utilizing a Futsal ball significantly accelerated the acquisition of passing skills (Oppici et al., 2018b) and led to a greater improvement in passing accuracy compared to training with a standard soccer ball in youth players (Caglayan et al., 2018). This effect was empirically observed even when testing occurred under the constraints of a larger soccer environment (Oppici et al., 2018a).
- **Dribbling Speed (Isolated Skill):** Conversely, short-term intervention studies found no statistically significant difference in the improvement of dribbling speed between groups training with the Futsal ball versus the soccer ball (Caglayan et al., 2018).

Perceptual-Motor and Cognitive Skills

Results concerning the transfer of speed-related and cognitive skills are highly divergent, necessitating careful consideration of the test context.

- **Reactive Agility and Decision Making Time (Field-Based):** Comparative cross-sectional studies on elite female players found that Futsal players performed significantly better in the Reactive Visual Stimuli Agility Test (RVS-T) and recorded significantly lower Decision Making Time (DMT) than their soccer counterparts (Benvenuti et al., 2010). Notably, no differences were observed in planned agility performance (Benvenuti et al., 2010).
- **Decision-Making Quality (Retrospective):** A retrospective study on Brazilian professional female players found a significant correlation between the accumulated hours of deliberate practice in Futsal during childhood (6–12 years) and better offensive tactical decision-making quality in adulthood (Machado et al., 2024). This relationship was primarily restricted to offensive situations involving the ball and near the ball (Machado et al., 2024).
- **Anticipation and Reaction Time (RT) (Laboratory/Isolated):** Analysis of game situations showed that the temporal dimension of a defender's response was crucial for dribbling success in soccer, linking success to the opponent's Reaction Time (RT). However, neither Reaction Time nor anticipation were found to be determinant factors for dribbling success in Futsal (Silva et al., 2022).

Contextual and Biomechanical Specificity

Research employing detailed kinematic and spatiotemporal analyses suggests that the key constraints governing success differ significantly between the two sports.

- **Dribbling Success Dynamics:** An analysis of attacker-defender interactions revealed that successful dribbles in Futsal were significantly associated with specific spatiotemporal coordination tendencies, namely the smallest interpersonal distances (ranging from 0.20 to 0.35 m) and the highest relative dribbling velocities (Silva et al., 2025). Crucially, the same spatiotemporal parameters that determined success in Futsal did not affect the dribbling outcomes in soccer (Silva et al., 2025).
- **Implications of Specificity:** This high contextual dependency suggests that Futsal demands players execute actions with extreme control and velocity when under immediate pressure in congested space, creating uncertainty for the defender due to proximity and speed (Silva et al., 2025). Conversely, success in soccer dribbling was found to be statistically related to delaying the defender's response (Silva et al., 2022).

Discussion

The primary purpose of this systematic review and meta-analysis was to quantitatively assess the transferability of technical, tactical, and perceptual-motor skills from Futsal to association Soccer, thereby examining the empirical support for the “donor sport” concept. The key quantitative finding of this study is the very large effect size ($SMD = 1.43$) favoring futsal-based interventions for passing accuracy. This synthesis suggests that Futsal acts as a potent catalyst for technical refinement, while the transfer of speed-related perceptual-motor skills is highly dependent on context and task specificity. Interpreting the findings through the Athletic Skills Model (ASM) clarifies why futsal may function as a donor sport for soccer. The very large effects observed for passing accuracy are consistent with the ASM’s emphasis on donor sports as constraint environments that promote adaptable movement solutions and perception–action coupling under time-pressure. Likewise, the divergent findings for time-based skills align with the ASM prediction that transfer depends on the representativeness of task constraints: field-based reactive assessments preserve informational variables and affordances closer to the donor sport, whereas isolated laboratory tasks may reduce ecological validity and therefore attenuate observable transfer.

Futsal as a Technical Development Accelerator

The robust quantitative evidence, derived partially from a randomized controlled trial (RCT) which achieved the highest methodological quality score (23/27) on the Downs & Black scale (Oppici et al., 2018b), confirms that Futsal practice promotes skill acquisition that successfully transfers to soccer performance (Oppici et al., 2018a, 2018b).

This significant technical transfer is attributed directly to the task constraints inherent to Futsal (Travassos et al., 2018):

- (1) **Ball Properties:** The Futsal ball possesses a lower coefficient of restitution (less bounce) (Peacock et al., 2017) and is considered easier to control than the soccer ball (Oppici et al., 2018b). This simplifies the kicking action, which in turn frees up the player’s attentional resources to focus on processing information related to the environment, thus supporting the passing action (Oppici et al., 2017, 2018b). Training with the Futsal ball was sufficient to accelerate the acquisition of passing skills and transfer them positively to performance with a standard soccer ball (Caglayan et al., 2018; Oppici et al., 2018b).
- (2) **Practice Volume and Intensity:** The constrained environment (smaller pitch size and fewer players) leads to players having more frequent contacts with the ball and operating under high opponent pressure and reduced time for action (Oppici et al., 2019; Travassos et al., 2017). This high intensity of skill execution promotes the development of superior technical control in confined spaces (Benvenuti et al., 2010).

This empirical evidence aligns with the perceptions of professional players: large majorities of U20 players recognized that Futsal practice contributed significantly to their passing ability (78.2%) and their dribbling/ball handling (71.3%) for soccer (Balzano et al., 2025).

The Paradox of Speed and Perceptual-Motor Specificity

The quantitative synthesis of time-based perceptual-motor skills (including agility and reaction time) resulted in a non-significant overall effect and very high statistical heterogeneity ($I^2 = 94.5\%$). This outcome highlights a critical distinction in skill transfer dependent on the ecological validity of the task (Andrew & Causer, 2022).

- **Ecological/Reactive Tasks:** Studies utilizing field-based reactive visual stimuli agility tests (RVS-T) found that Futsal players were significantly faster in reactive agility and displayed lower Decision-Making Time (DMT) compared to their soccer counterparts (Benvenuti et al., 2010; Naser et al., 2017). This superiority disappeared in tests measuring planned agility (PVS-T) (Benvenuti et al., 2010). This indicates that the Futsal environment sustains the development of quick cognitive and coordinative skills needed to respond rapidly to unexpected stimuli (Benvenuti et al., 2010; Naser et al., 2017).
- **Isolated/Non-Ecological Tasks:** Conversely, investigations into anticipation using video-based temporal occlusion tests (e.g., penalty kicks) found no evidence of transfer from similar invasion sports (Andrew & Causer, 2022). For complex perceptual-cognitive tasks like anticipating penalty kicks, the necessary processes were found to be sport-specific (Andrew & Causer, 2022).

The Futsal advantage, therefore, seems to be rooted in improving the perception-action coupling—the ability to perceive environmental information (affordances) and execute quick, functional actions under pressure (Oppici et al., 2018a)—rather than enhancing general visual processing speed (Oppici et al., 2017).

Talent Development Pathways and Cognitive Investment

Retrospective studies confirm the developmental importance of Futsal, particularly for cognitive skills (Machado et al., 2023, 2024). The accumulation of deliberate practice in Futsal during childhood (6–12 years) and early adolescence (13–15 years) correlated positively with the quality of offensive decision-making skills in adult professional female soccer players (Machado et al., 2023, 2024). Specifically, players with higher cognitive performance had accumulated more hours of deliberate practice in Futsal and higher total practice hours during early adolescence (Machado et al., 2024).

These findings support the Specialized Diversification pathway, which advocates for involvement in diverse contexts related to the primary sport (including Futsal) during childhood, followed by a shift toward structured practice during specialization

(Sieghartsleitner et al., 2018). The fact that elite players accumulated significantly more time in Futsal practice aligns with the notion that the Futsal environment requires players to “think more” (Cunha Neto, 2013), fostering cognitive skills deemed essential for modern soccer (Machado et al., 2024).

Dribbling Performance and Biomechanical Specificity

The transfer of dribbling skills shows high specificity, influenced heavily by the distinct spatiotemporal constraints of each sport:

- **Futsal Dribbling Dynamics:** Successful Futsal dribbles are associated with precise spatiotemporal coordination in the dyadic interaction, occurring when interpersonal distances are smallest (e.g., 0.20 to 0.35 m) and the dribbling velocities are highest (Silva et al., 2025). This high-pressure environment necessitates refined control.

- **Soccer Dribbling Dynamics:** Conversely, the spatial and velocity factors that constrain Futsal dribbling outcomes were not determinant for successful dribbles in the analogous soccer scenarios (Silva et al., 2025). Instead, success in soccer dribbling was found to be statistically related to the defender’s Reaction Time (RT), suggesting that the attacker aims to delay the defender’s response rather than relying primarily on close-quarters skill (Silva et al., 2022).

Furthermore, biomechanical studies highlight how Futsal kicks differ based on task complexity. Kicking a rolling ball (a more complex and ecologically valid Futsal action) showed higher asymmetry for accuracy in the non-dominant limb compared to kicking a stationary ball, indicating that Futsal’s demands magnify existing bilateral differences, largely due to specific angular adjustments required in the hip joint (Barbieri et al., 2015). This suggests that coaches should emphasize practice with the non-dominant limb to mitigate these asymmetries (Barbieri et al., 2015).

Strengths and Limitations

This systematic review and meta-analysis presents inherent limitations that warrant consideration. Methodologically, title/abstract screening and full-text eligibility assessment were independently performed by two reviewers; however, inter-rater reliability statistics were not computed. Furthermore, the quantitative synthesis is based on a limited number of included studies ($N = 7$) and demonstrated high statistical heterogeneity (e.g., I^2 values exceeding 86% in both key quantitative analyses), suggesting significant variance across experimental designs. The conclusions regarding transfer are also limited by the prevalence of cross-sectional or observational designs in the primary literature, hindering the establishment of direct causality (Machado et al., 2023, 2024). Finally, studies relying on retrospective self-reported data carry the inherent risk of recall bias (Machado et al., 2023, 2024).

Conclusion and Practical Applications

The body of current scientific evidence rigorously validates the “donor sport” concept, particularly in the domain of technical passing proficiency. Futsal provides a complementary learning environment characterized by unique constraints that accelerate technical acquisition and refine quick, context-dependent cognitive skills.

Practical Recommendations

- (1) Technical Training: Coaches should endorse the systematic integration of the Futsal ball and Futsal-like constraints (e.g., small-sided games) in youth development programs to maximize the acquisition and positive transfer of passing accuracy (Caglayan et al., 2018; Oppici et al., 2018b).
- (2) Talent Development: The evidence supports the Specialized Diversification pathway, emphasizing that a high volume of deliberate practice in Futsal, especially in childhood and early adolescence, promotes the development of high-quality offensive decision-making skills in adult soccer (Machado et al., 2023, 2024).
- (3) Assessment: Reactive agility tests conducted under ecological, field-based conditions (RVS-T) can be reliable measures to identify players with superior, transferred decision-making and cognitive speed (Benvenuti et al., 2010).

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Data Availability Statement

The search outputs (RIS exports), screening logs (HTML), and figures supporting this review are available in an Open Science Framework (OSF) repository and can be visualized in the next URL: https://osf.io/ejbwc/overview?view_only=2a033a126a9f49f78015dd286d544ab9.

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